AI SOLUTION FOR E-COMMERCE & LOGISTICS

Catch damaged parcels before they leave the warehouse

AI-based parcel damage detection for Takealot

THE PROBLEM

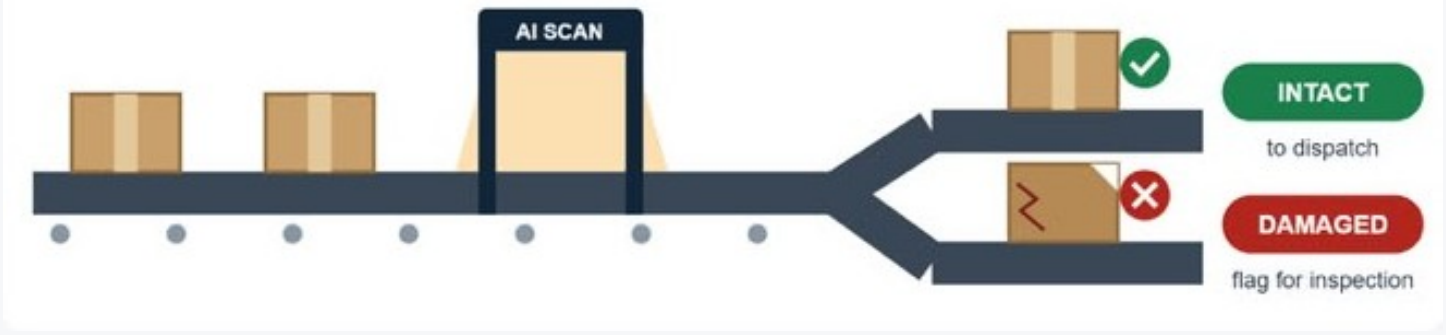
Damaged parcels ship unnoticed. The damage is found only after delivery, causing refunds, replacements and unhappy customers.

THE SOLUTION

An AI camera scans each parcel on the conveyor and labels it Damaged or Intact with a confidence score. Staff repackage flagged parcels before dispatch.

HOW IT WORKS





1. Arrive.
Parcels reach the belt before dispatch.

2. Scan.
The camera checks for dents, tears, holes and crushed corners.

3. Sort.
Intact parcels dispatch; damaged ones are flagged.

Concept illustration; confidence values shown are illustrative. Our working prototype classifies a photo of a parcel as Damaged or Intact.

RESULTS SO FAR

~91%
best validation accuracy (100 images)

27/27
unseen test images classified correctly

349
training images in 2 classes

The test set is small, so we treat ~91% as our reliable figure and will confirm it on more data. A basic CNN built from scratch scored only 52%; transfer learning made the difference.

UNDER THE HOOD

KEY FEATURES

- Damaged / Intact result with confidence score
- Photo upload or live camera
- Simple web app for warehouse staff

TECHNOLOGY

- Python, TensorFlow / Keras
- MobileNetV2 transfer learning
- Streamlit web app

PLANNED NEXT

- Track damage rates over time
- Saved record of every parcel

BENEFITS

- **Fewer damaged parcels dispatched**
- **Lower refund and return costs**
- **Faster, more consistent inspection**